Housing Data PrepFlow

HOUSING DATA PREPFLOW

AMES HOUSING DATASET ANALYSIS

Houses	Quantity Sold	Avg Price
2,930	82	$180K

2. Cleaning Summary

2.1 Missing Values

Column	Missing %	Solution	Reason
Alley	93%	"None"	No alley access
Lot Frontage	17%	Median	Robust to outliers

2.2 Duplicates

Duplicates Removed	Before Rows	After Rows
9	2,930	2,921

2.3 Outliers

- Gr Liv Area: Capped at 99th percentile
- SalePrice: Visual inspection via Boxplot

4. Key Findings

4.1 Three Most Interesting Discoveries

- **Quality is the Strongest Price Predictor**

Correlation analysis revealed that Overall Qual has the highest correlation (0.79) with SalePrice. This suggests that subjective quality ratings are more predictive of price than pure size measurements.

- **Living Area Shows Diminishing Returns**

While larger homes generally command higher prices, the relationship is not perfectly linear. Very large homes (>3000 sq ft) don't always proportionally increase in value.

- **Price Distribution is Right-Skewed**

The SalePrice distribution shows clear right-skewness with mean ($180,796) significantly higher than median ($160,000). This indicates high-value outliers that pull the average up.

1. Introduction

The Ames Housing Dataset contains information about residential properties sold in Ames, Iowa (2006-2010). It includes 82 features covering: size, quality, location, and sale price.

3. Feature Engineering

3.1 New Features Created

New Feature	Formula	Purpose
Price_per_Sqft	SalePrice / Gr Liv Area	Price normalized by size
TotalArea	Gr Liv Area + Total Bsmt SF	Combined living space
quality_x_area	Overall Qual × Gr Liv Area	Capture interaction
House Age	2026 - Year Built	Numeric age
Age Group	New/Recent/Old	Age categories
Gr Liv Area Log	log(Gr Liv Area + 1)	Reduce skewness

3.2 Encoding Techniques

- **One-Hot Encoding**

Applied to categorical variables with limited categories:
1- Lot Shape: IR1, IR2, IR3, Reg
2- House Style: 1Story, 2Story, 1.5Fin, etc.

- **Ordinal Encoding**

Applied to naturally ordered categories:
Kitchen Qual: Po=1, Fa=2, TA=3, Gd=4, Ex=5

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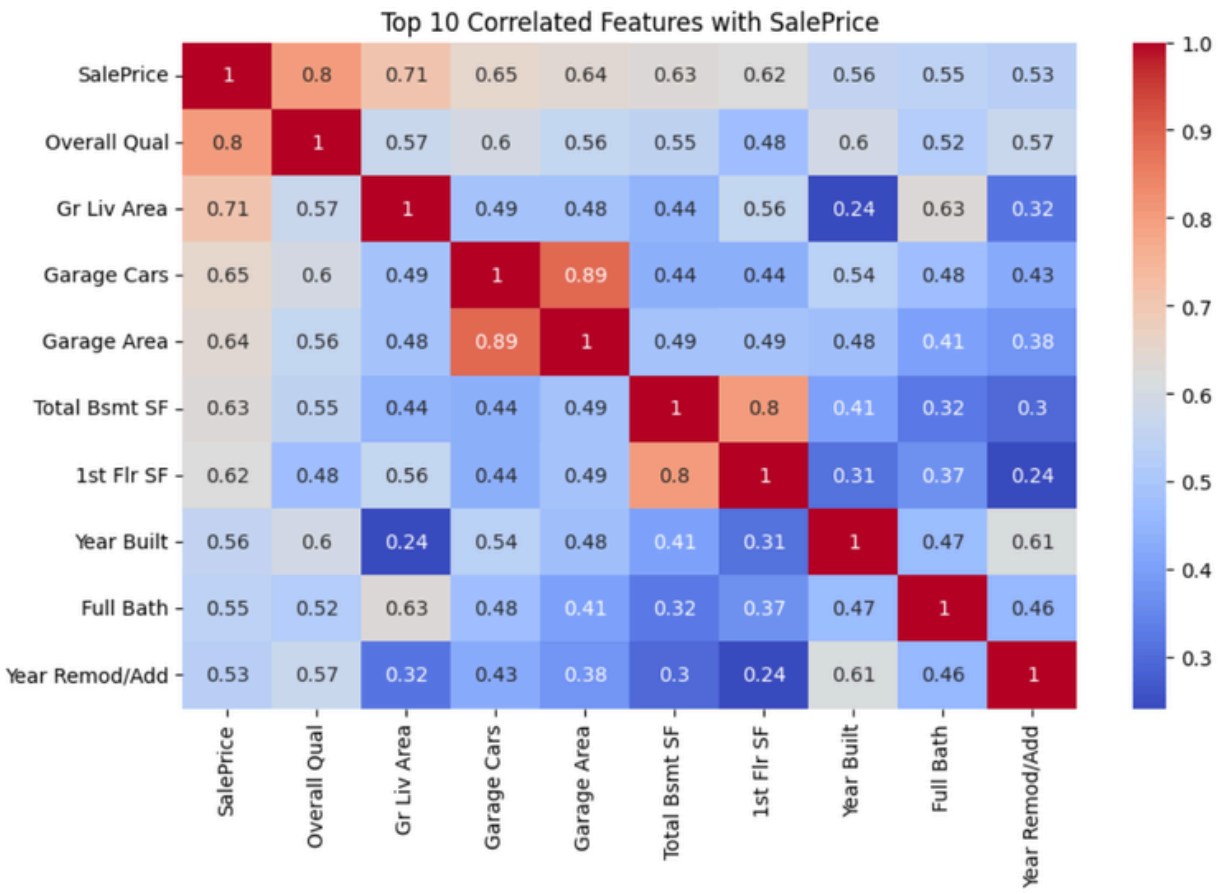
4.2 Best Chart -Correlation Heatmap



Insight: Overall Quality and Gr Liv Area appear among the strongest predictors of SalePrice.

4.3 Top Correlated Features

- Overall Quality is the strongest price predictor (0.79)
- Gr Liv Area strongly correlates with price (0.71)
- SalePrice is right-skewed
- The best chart is the Correlation Heatmap showing all top predictors

Feature	Correlation
Overall Qual	0.79
Gr Liv Area	0.71
Garage Cars	0.64
Total Bsmt SF	0.63
1st Flr SF	0.61

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MATHEMATICAL ANALYSIS

Count (n)	Sum (Σx)	Std Dev (σ)	Mean (μ)
2,921	$527.7M	$79,886	$180,796

5.Standard Deviation Formula

Formula:

$$\sigma = \sqrt{[\Sigma(x - \mu)^2 \div n]}$$

Where: x = each value, μ = mean (180,796), n = total count (2,921)

5.1Cosine Similarity Analysis

House	Price	Area	Quality
Most Expensive	$755,000	4,342 sq ft	10
Least Expensive	$12,789	480 sq ft	1

5.2 Probability Analysis

Question: What is the probability that a house has price > $200,000 if Quality ≥ 7?

High Quality Houses	Above $200K	Probability
1,090	742	68.07%

6. Future Work

6.1 Short-term Improvements (1-2 weeks)

- Advanced Feature Engineering: Create polynomial features
- Model Development: Train baseline models (Linear Regression, Random Forest)
- Cross-validation: Implement robust evaluation

6.2 Short-term Improvements (1-2 weeks)

- Advanced Analytics: Time series models
- Geospatial Analysis: Neighborhood clustering
- Deployment: Deploy to cloud platform

7. Conclusion

This comprehensive analysis of the Ames Housing Dataset has revealed valuable insights into the factors influencing house prices. Key takeaways include:

Quality matters most: Overall quality rating is the strongest predictor

Size is important but not everything: Living area correlates strongly but with diminishing returns

Data quality is critical: Proper cleaning impacts analysis accuracy

Feature engineering adds value: Created features provide additional predictive power

This analysis provides a solid foundation for building predictive models and understanding the Ames housing market.

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